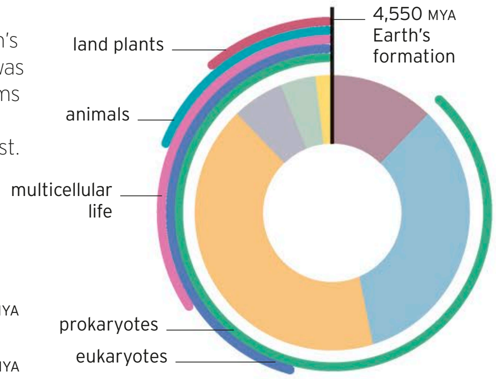


▷ **GEOLOGICAL TIME**
Over the enormous span of Earth's history, times during which there was no life or only the simplest life forms have lasted longer than the eras where multicellular life-forms exist.

KEY
- Cenozoic 66–0 MYA
- Mesozoic 252–66 MYA
- Paleozoic 541–252 MYA
- Proterozoic 2,500–541 MYA
- Archean 4,000–2,500 MYA
- Hadean 4,550–4,000 MYA

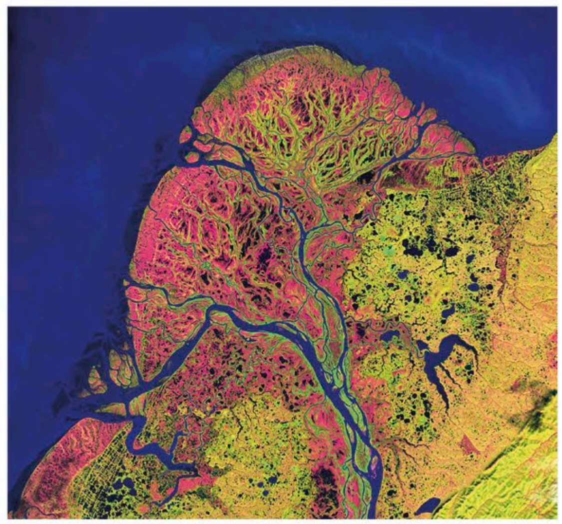
intervals called eons—the Hadean followed by the Archean and Proterozoic eons. It is now known that simple organisms called prokaryotes (single-celled bacteria-like organisms) first evolved during the Hadean. Some 2 billion years then passed before more complex single-celled organisms called eukaryotes evolved (see pp.16–17).

Agents of change

Geologists have argued for centuries about the nature of the processes, operating through Earth's past, that have brought about the varied landscapes seen today. In the late 18th century, support grew for the argument that Earth's features were mostly formed by slow, gradual, and ongoing changes over time. It was clear, for example, that erosion had shaped the planet everywhere. This view contrasted with the opposing doctrine, which proposed that a series of cataclysms was responsible for most of the Earth's surface features. Today, it is agreed that most landforms on Earth now are the result of gradual processes. Nevertheless, there have been catastrophes: for example, an asteroid or comet impact about 67 million years ago (MYA) is thought to have caused fires, massive earthquakes, worldwide darkness, and possibly wiped out the dinosaurs.

Changes in climate and sea level

Earth's climate has changed quite drastically over long periods of time. At one extreme, there have been times when there were no ice caps and temperate deciduous forests grew at the poles. At the other, it seems probable that Earth was at least once completely frozen over. Similarly, sea level has fluctuated dramatically. For much of the past 540 million years, it has been higher than it is today. But, for the last 2.5 million years or so, sea level has mostly been lower, since during this time Earth has been in an ice age with much of its water locked up in polar ice sheets. Over the past 17,000 years or so, Earth has been in a warming period within this ice age, called an interglacial, and sea level has risen again. Since the late 18th century, the planet has warmed by around 1.8°F (1°C), due to increased atmospheric carbon dioxide, and sea level has risen by about 12 in (30 cm).



◁ **SLOW BUILDUP**
Deltas, such as the Yukon Delta seen here, are examples of a landform built up slowly over a long period of time.

▽ **ERODED LANDFORM**
Horseshoe Bend in Arizona exhibits the effects of 75 million years of erosion. The rock layers in its walls range from around 180 to 200 million years old.